Ethanol Blending in Gasoline - Chile

July 2026

U.S. GRAINS & BIOPRODUCTS COUNCIL

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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

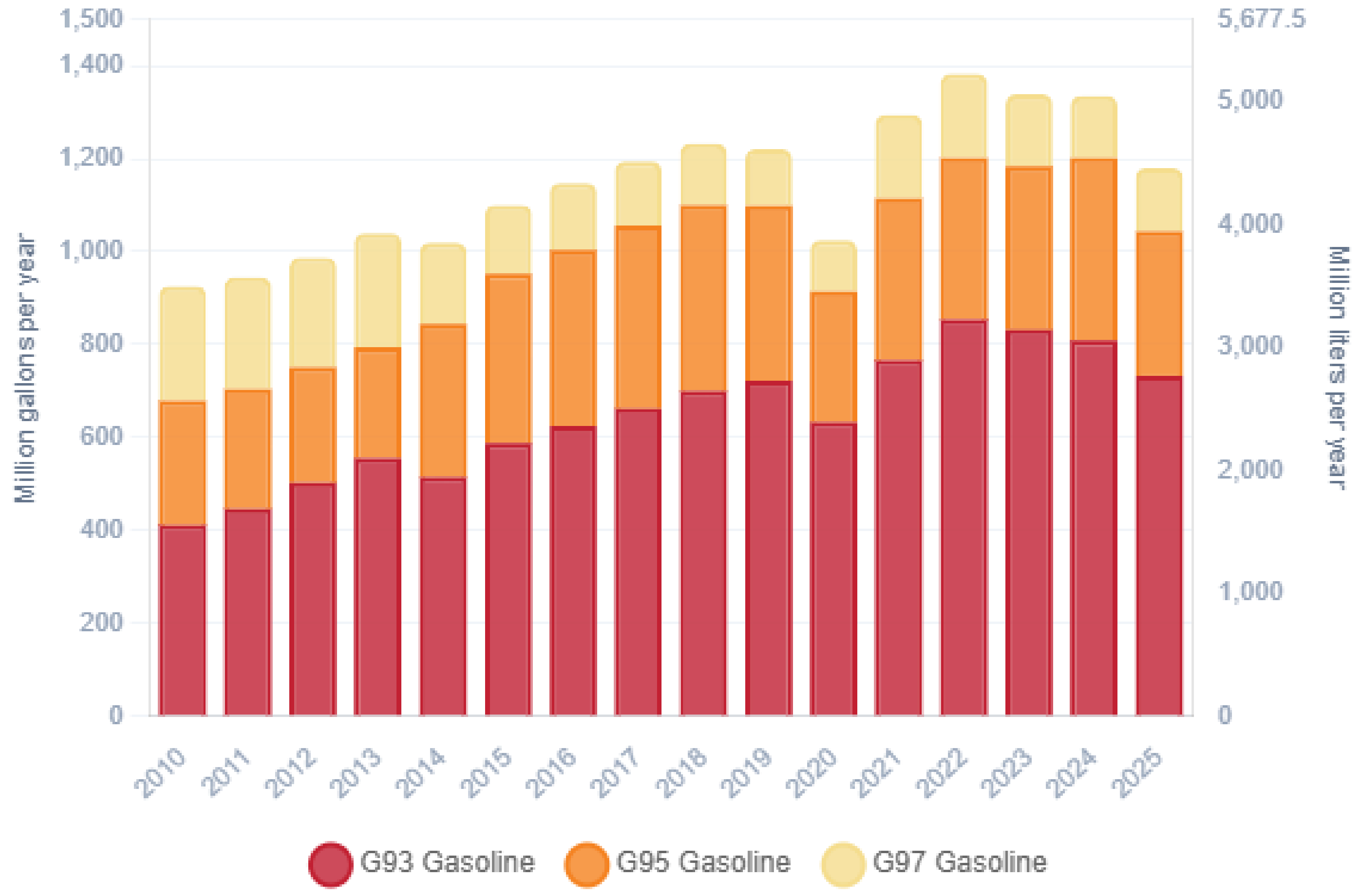
Ethanol Blending in Gasoline - Chile

Chile's gasoline consumption is currently over 1,180 million gallons per year (4,460 million liters) distributed in three grades: RON 93 (AKI 87), RON 95 (AKI 88) and RON 97 (AKI 92). There is a more stringent quality specification for Santiago Metropolitan Area. In 2025, the market share of RON 93 gasoline was 62%. Government company ENAP produces 77% of the total demand (CNE). Gasoline imports made in the country come mainly from the United States. Refineries only produce RON 93 and RON 97, gasoline RON 95 is blended from these grades in gas stations.

Ethanol can be blended up to 5% v/v (Decreto Supremo 56) but it is not used because of regulatory barriers. Chile does not produce or imports ethanol. The use of ethanol is being analyzed to achieve short term goals in the national economy decarbonization program.

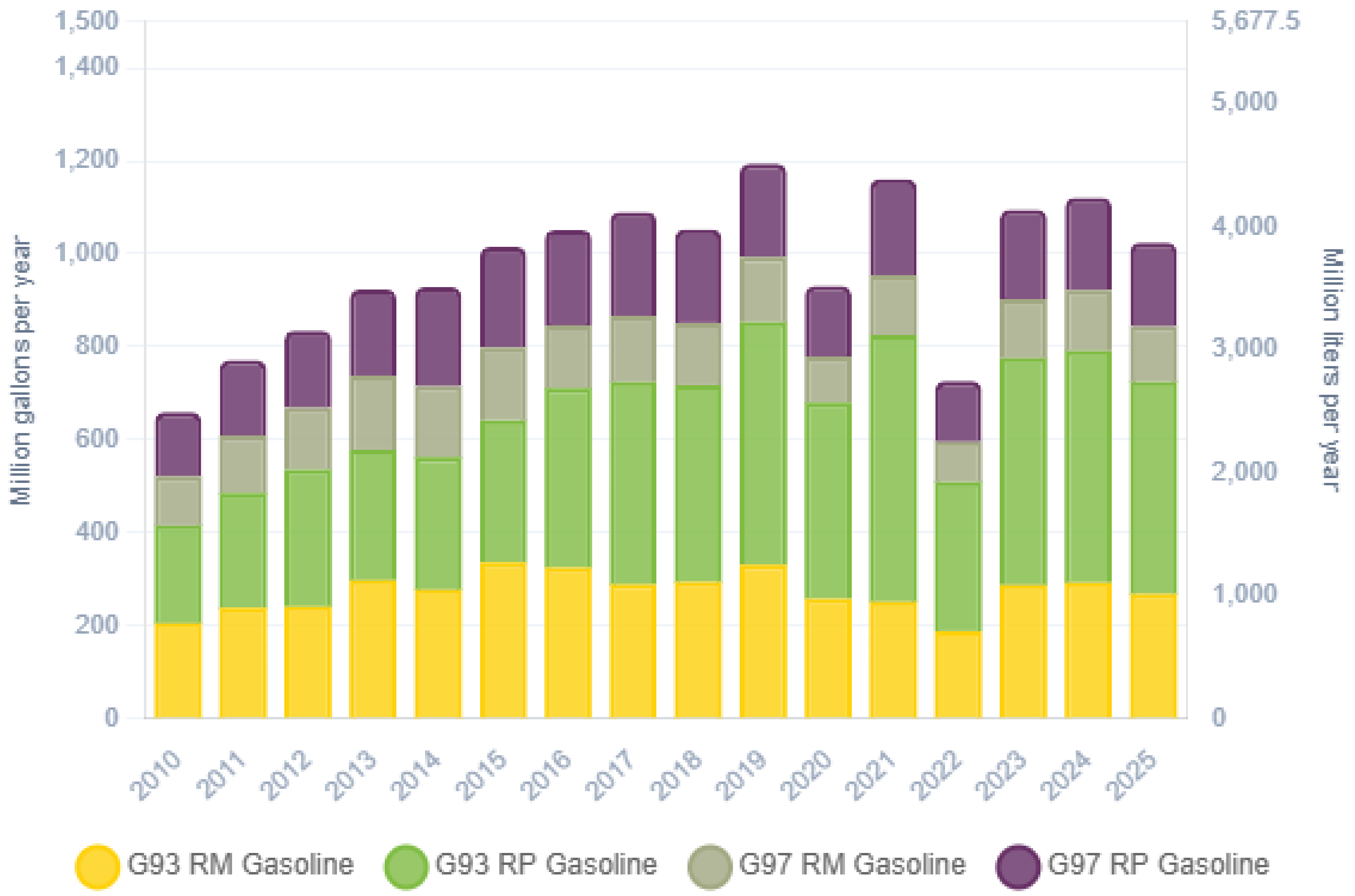
Source: CNE, SEC, 2025

Gasoline Demand in Chile



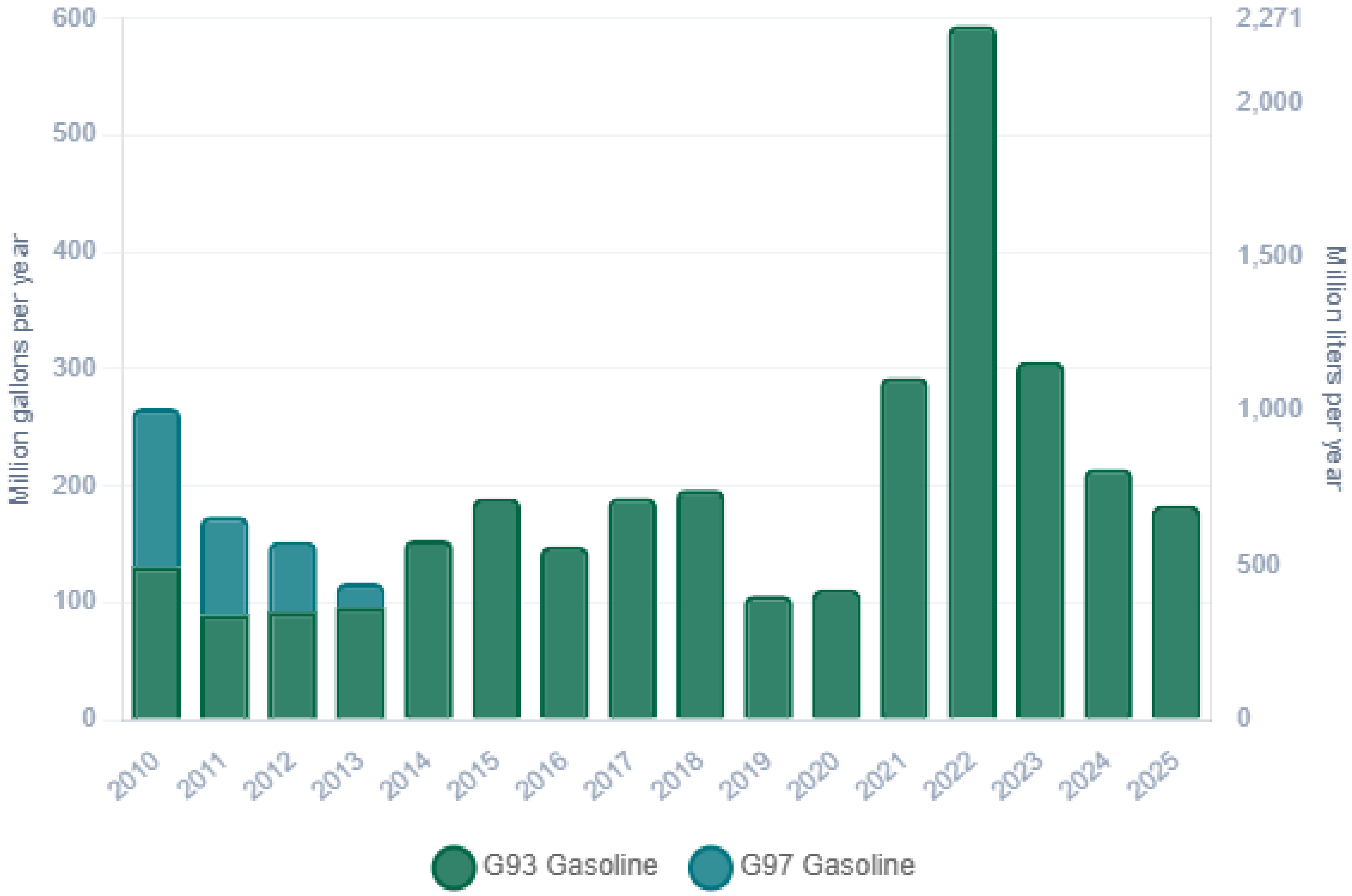
Fuente: CNE, 2025

Gasoline Production in Chile



Fuente: CNE, 2025

Gasoline Imports in Chile



Fuente: CNE, 2025

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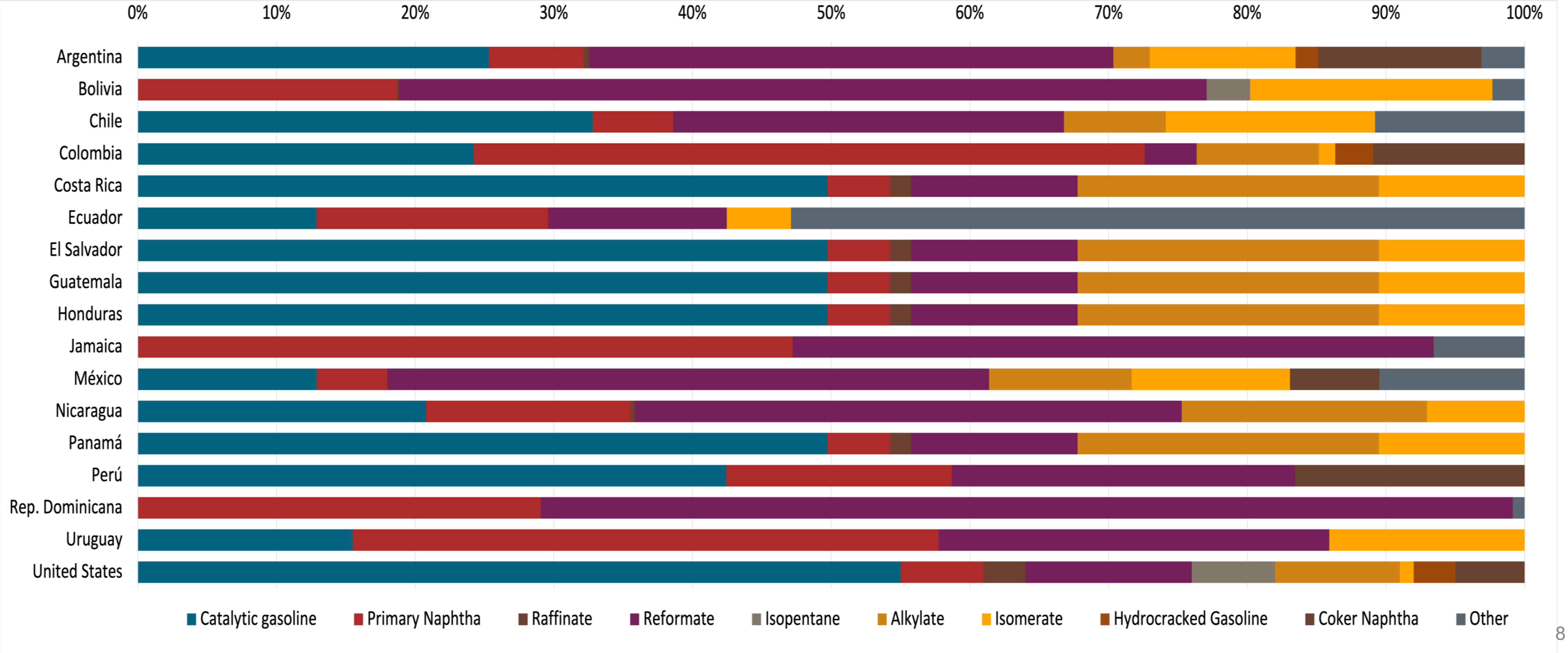
Gasoline Quality in Chile

Name	PPDA DS31	PPDA DS31	DS 60	DS 60	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017	2017	2012	2012	2017			
Applicability	Metropolitan Region	Metropolitan Region	Rest of the country	Rest of the country	All countries			
Selected Grade	G93	G97	G93	G97	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	12 %v/v	12 %v/v	20 %v/v	20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	93	97	93	97	> 95	> 95	> 98	> 98
MON	Report	Report	Report	Report	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2 %m/m	< 2 %m/m	< 2 %m/m	< 2 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 55 kPa	<> 55 kPa	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 69 kPa	<> 69 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: ENAP, 2022

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



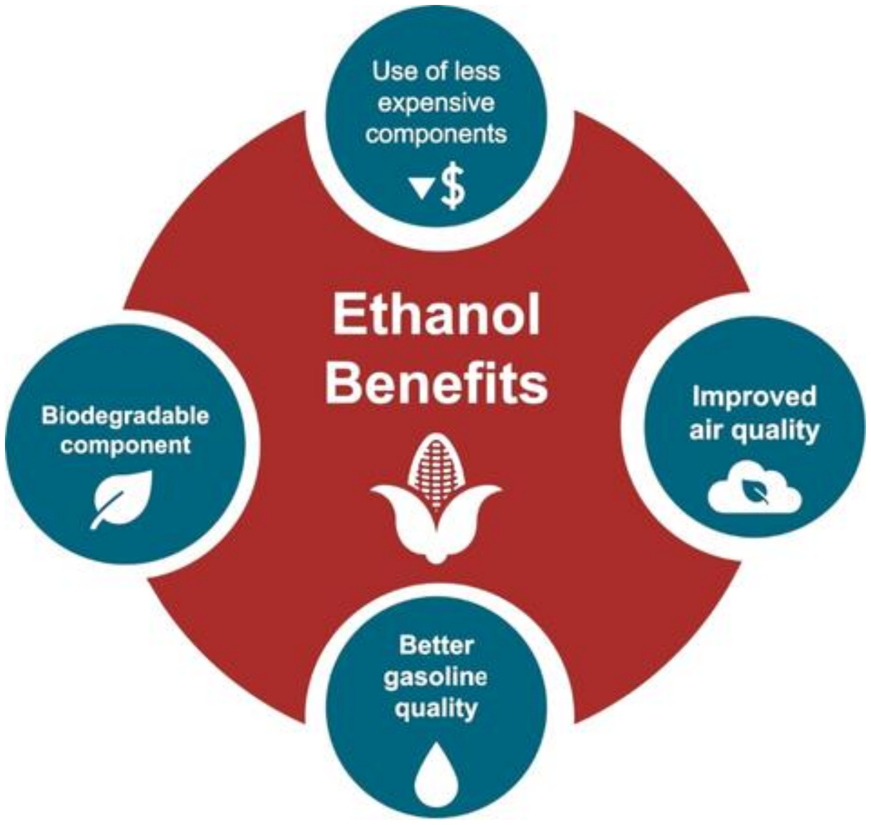
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

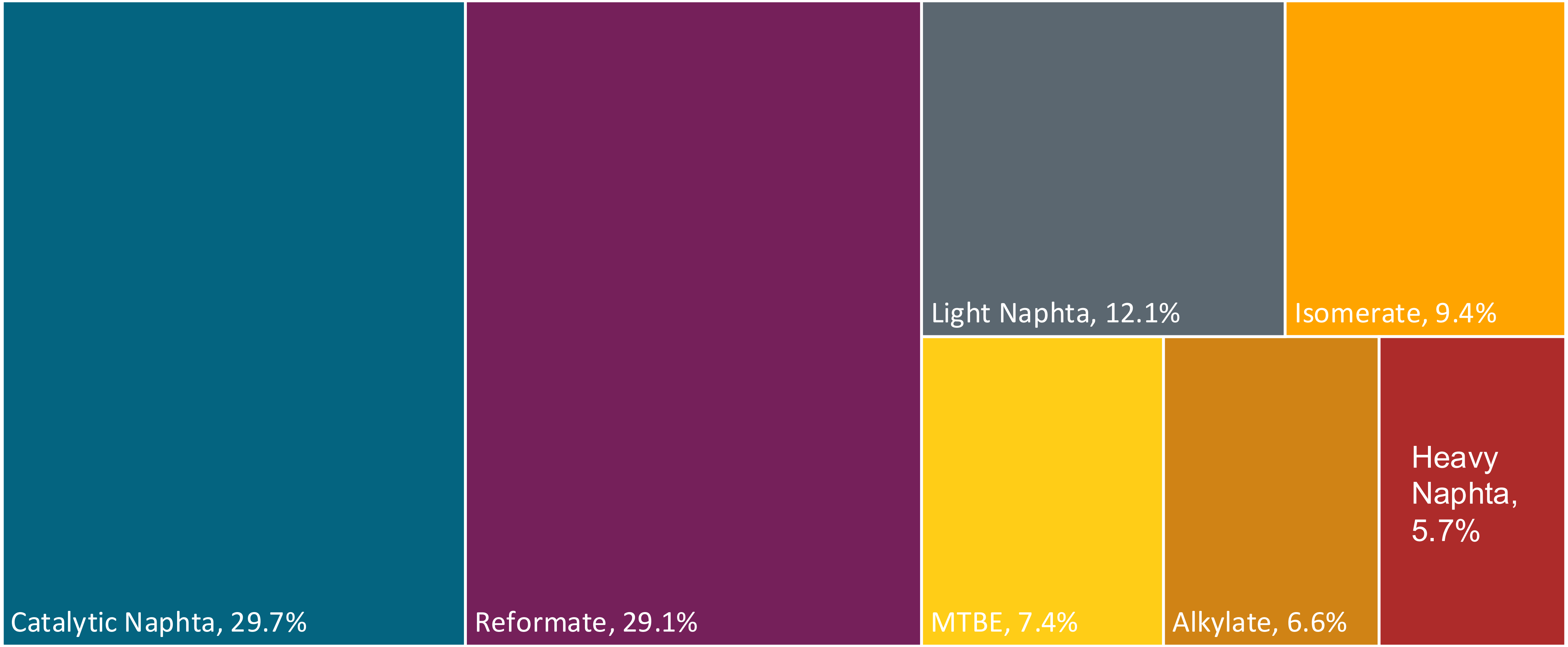
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

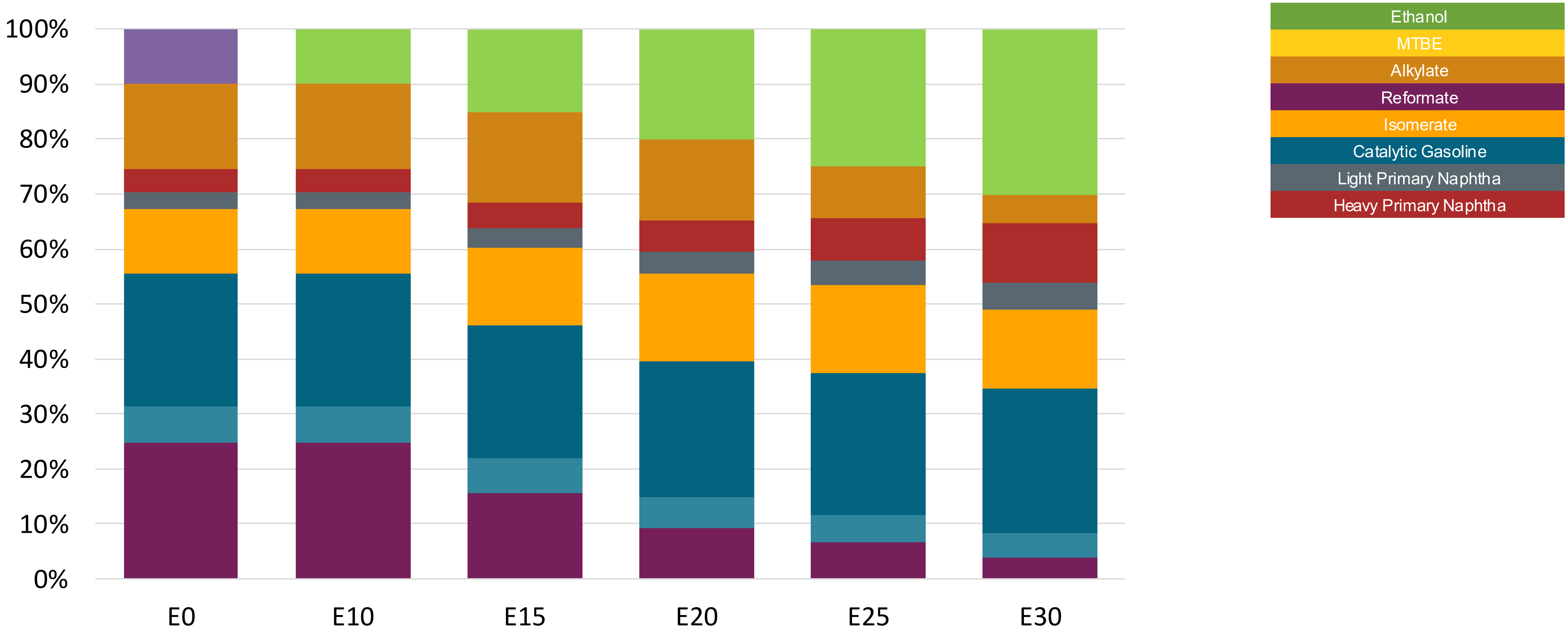


Available Blending Components



Fuente: ENAP, 2025

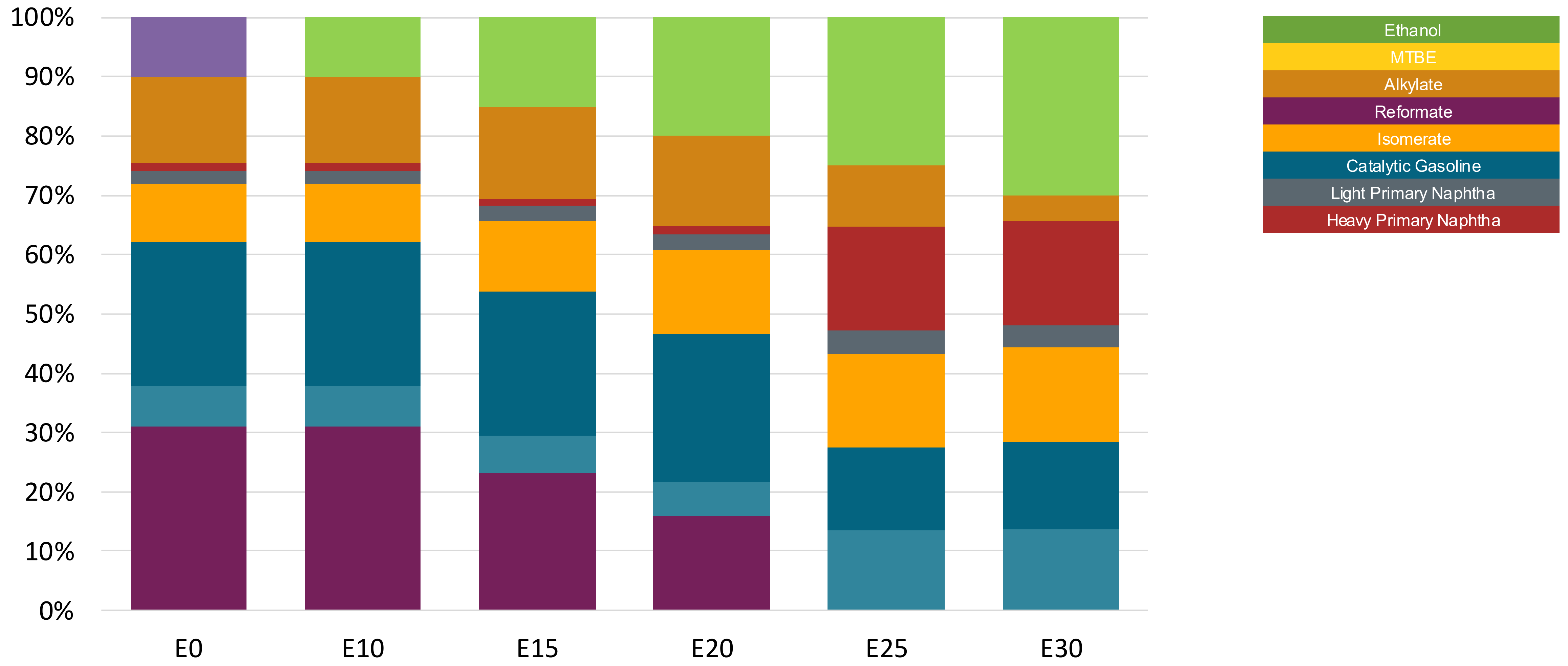
Chile – G93 RP Aconcagua – Constant Octane



Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.390	$2.308	$2.219	$2.131	$2.039	$1.949

Source: Faro90, 2025

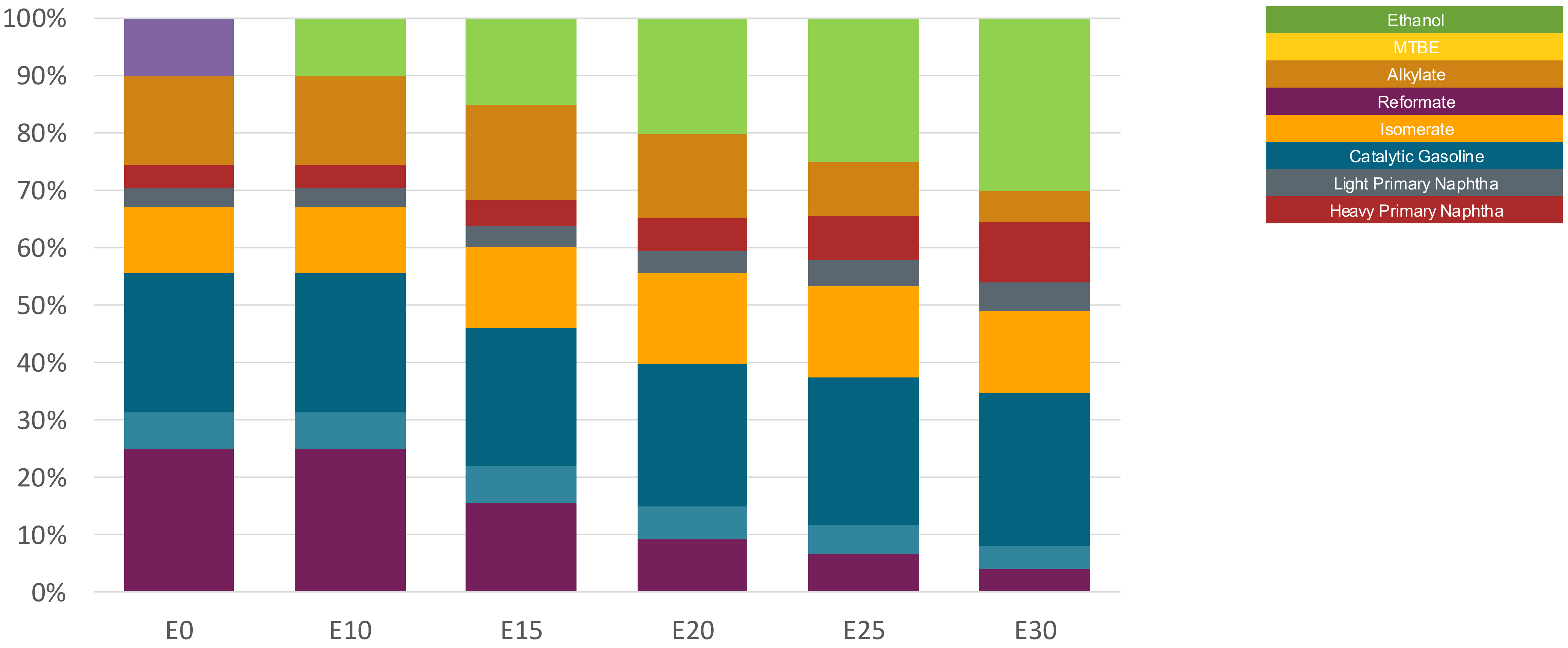
Chile – G97 RP Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.516	$2.434	$2.356	$2.276	$1.922	1.867

Source: Faro90, 2025

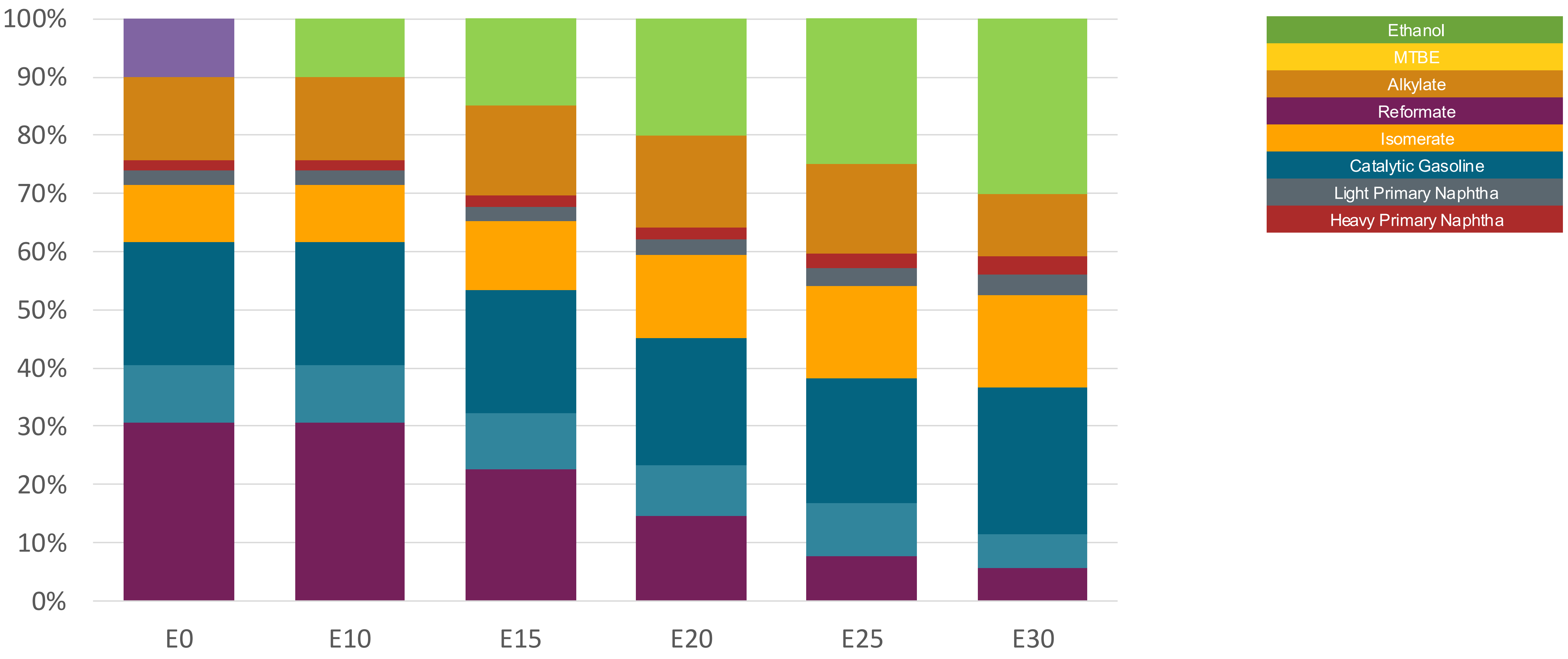
Chile – G93 RM Aconcagua – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.390	$2.308	$2.219	$2.131	$2.039	$1.949

Source: Faro90, 2025

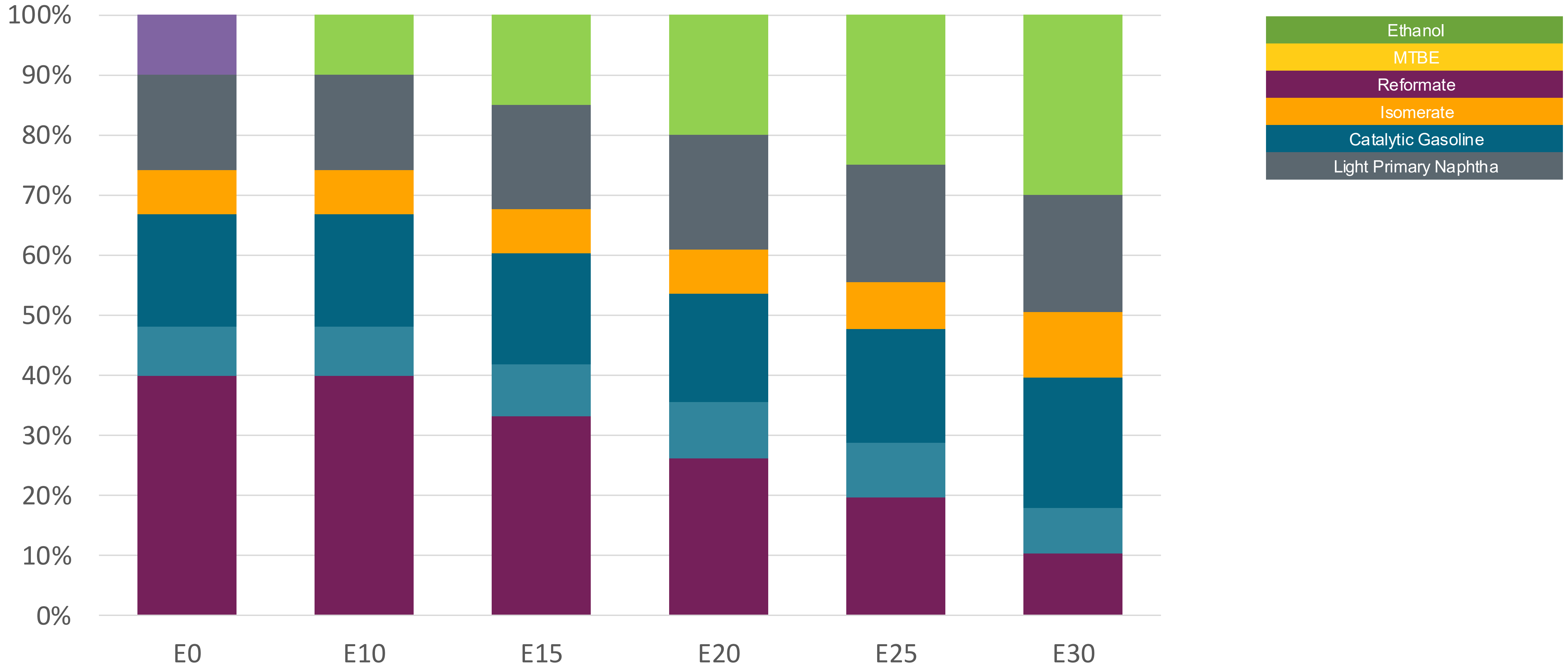
Chile – G97 RM Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.50	$2.42	$2.34	$2.26	$2.18	$2.11

Source: Faro90, 2025

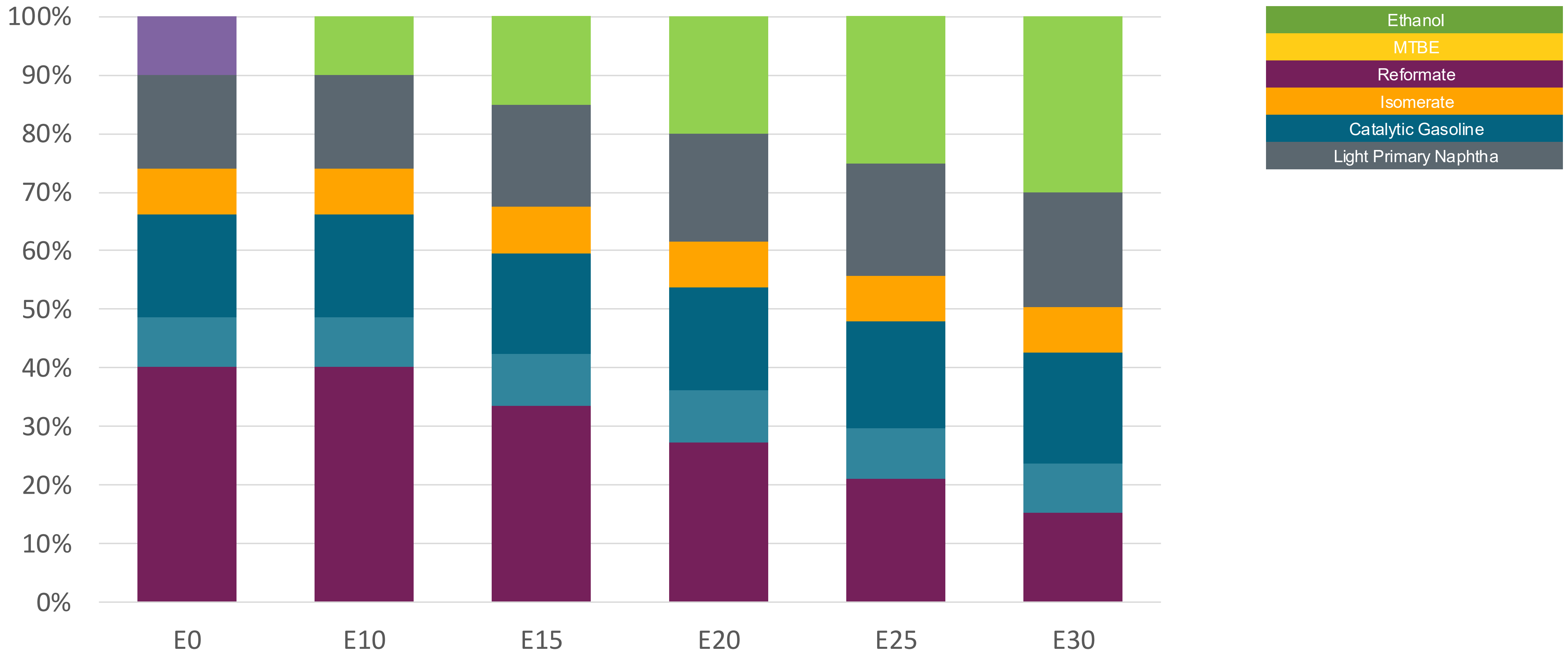
Chile – G93 RP Biobío – Constant Octane



Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US$/gal)	$ 2.25	$ 2.17	$ 2.09	$ 2.00	$ 1.93	$ 1.84

Source: Faro90, 2025

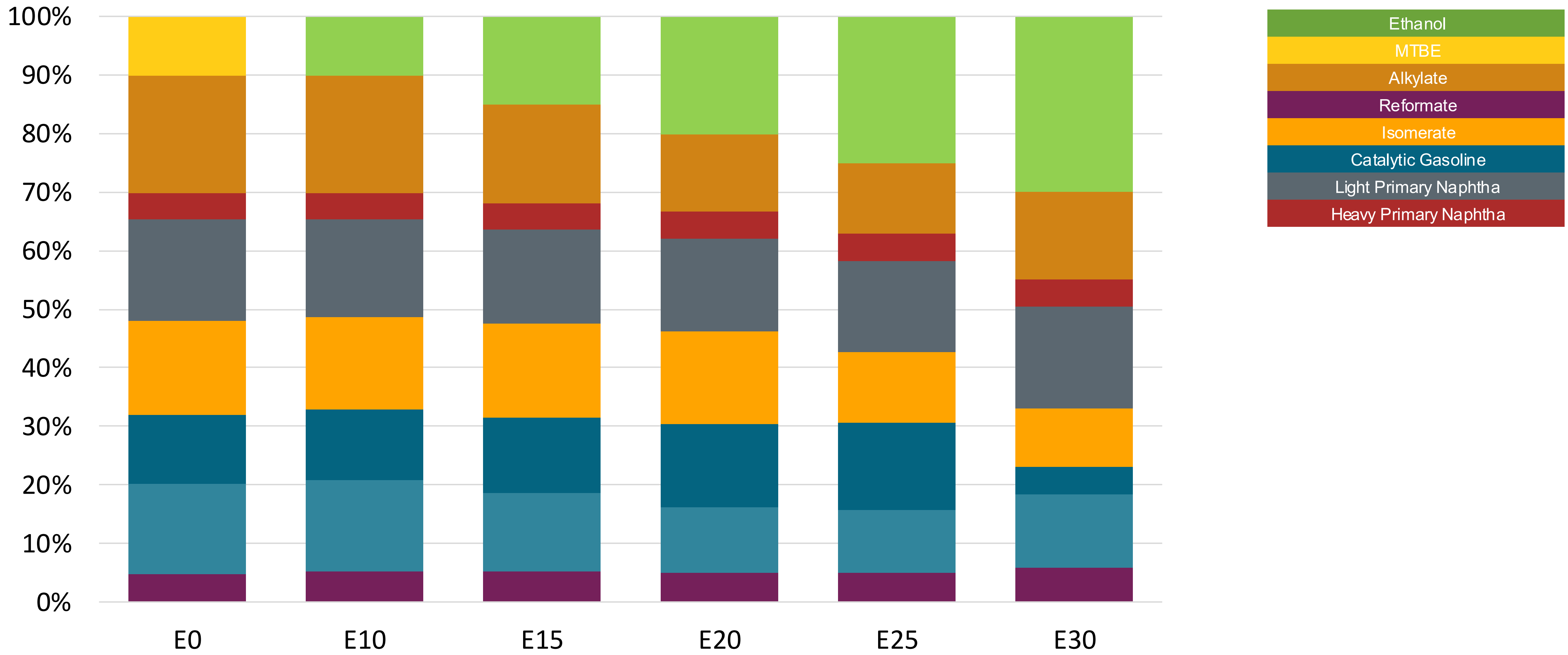
Chile – G97 RP Biobío – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.34	$ 2.26	$ 2.18	$ 2.10	$ 2.02	$ 1.96

Source: Faro90, 2025

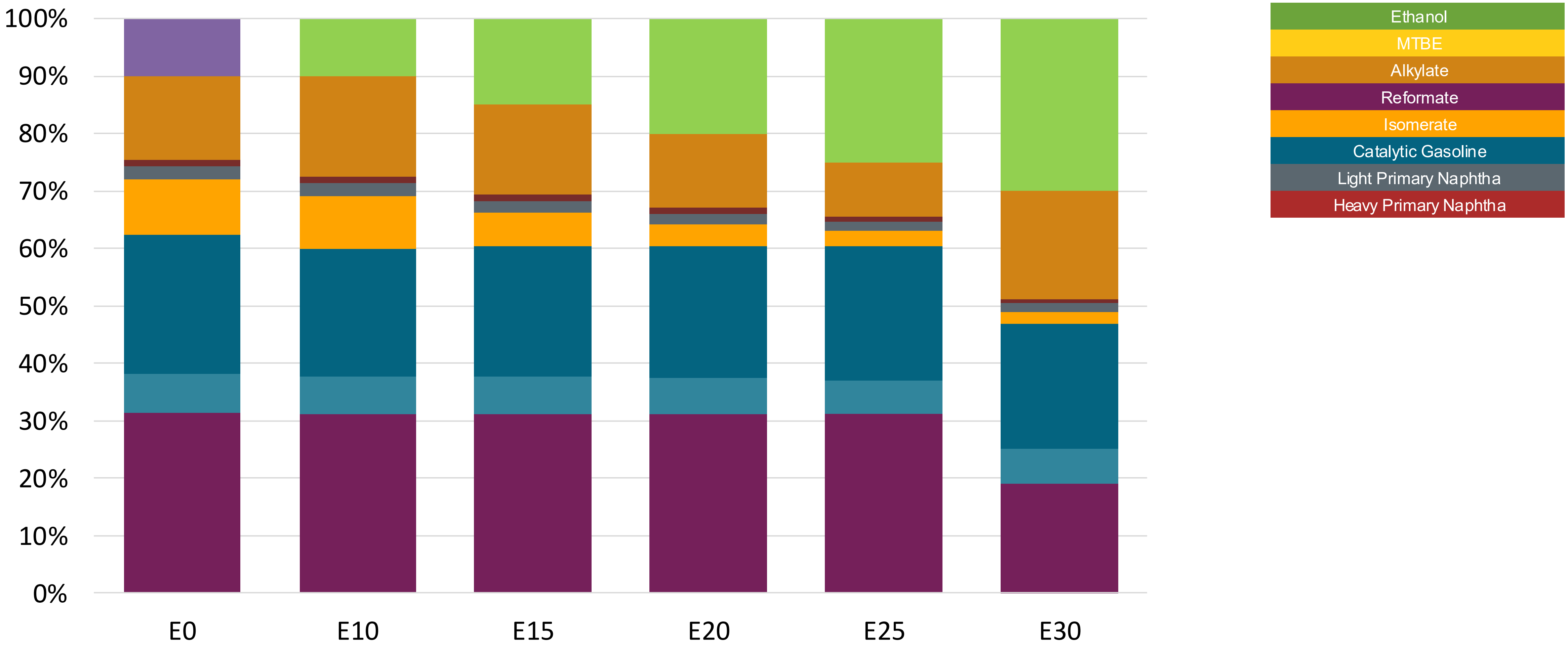
Chile – G93 RP Aconcagua – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	93.0	95.1	97.5	100.1	102.0	104.3
Price (US$/gal)	$ 2.19	$ 2.19	$ 2.19	$ 2.19	$ 2.19	$ 2.19

Source: Faro90, 2025

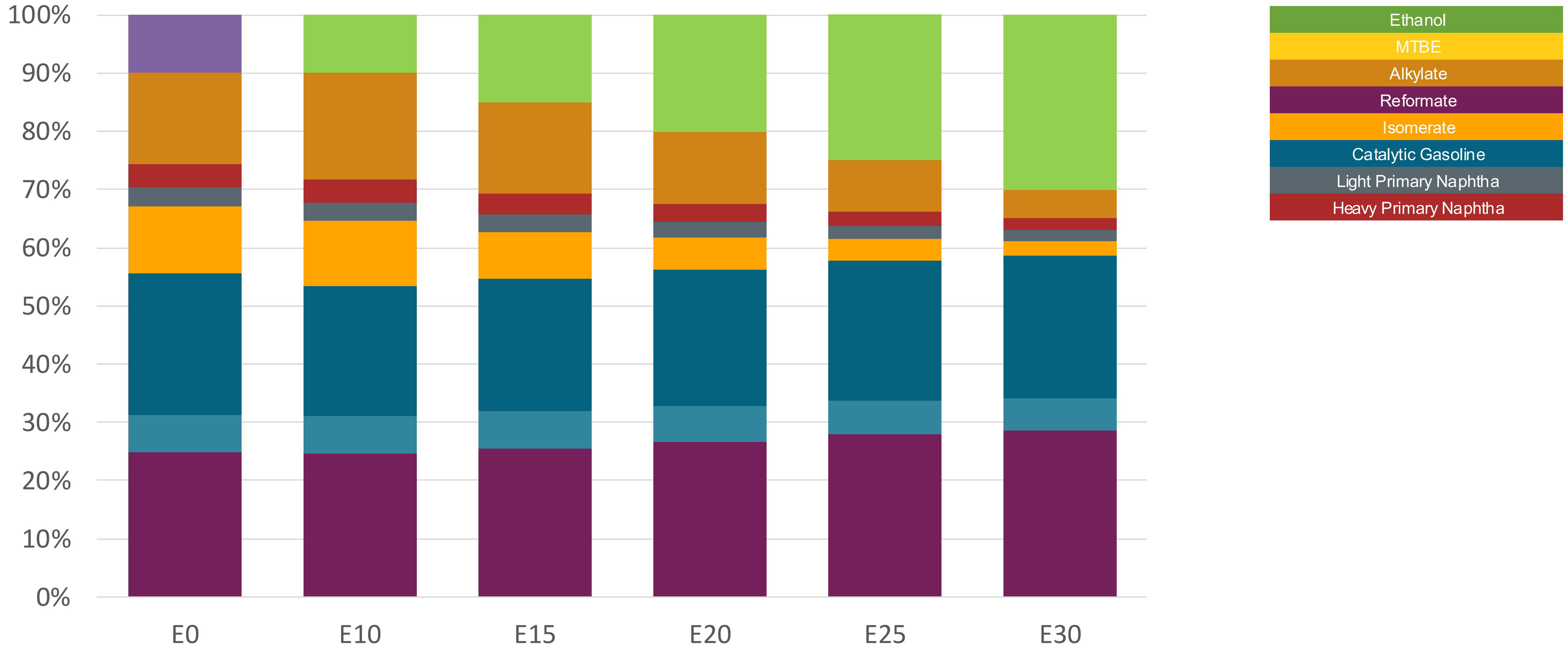
Chile – G97 RP Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.4
Price (US$/gal)	$2.52	$2.52	$2.52	$2.52	$2.52	$2.52

Source: Faro90, 2025

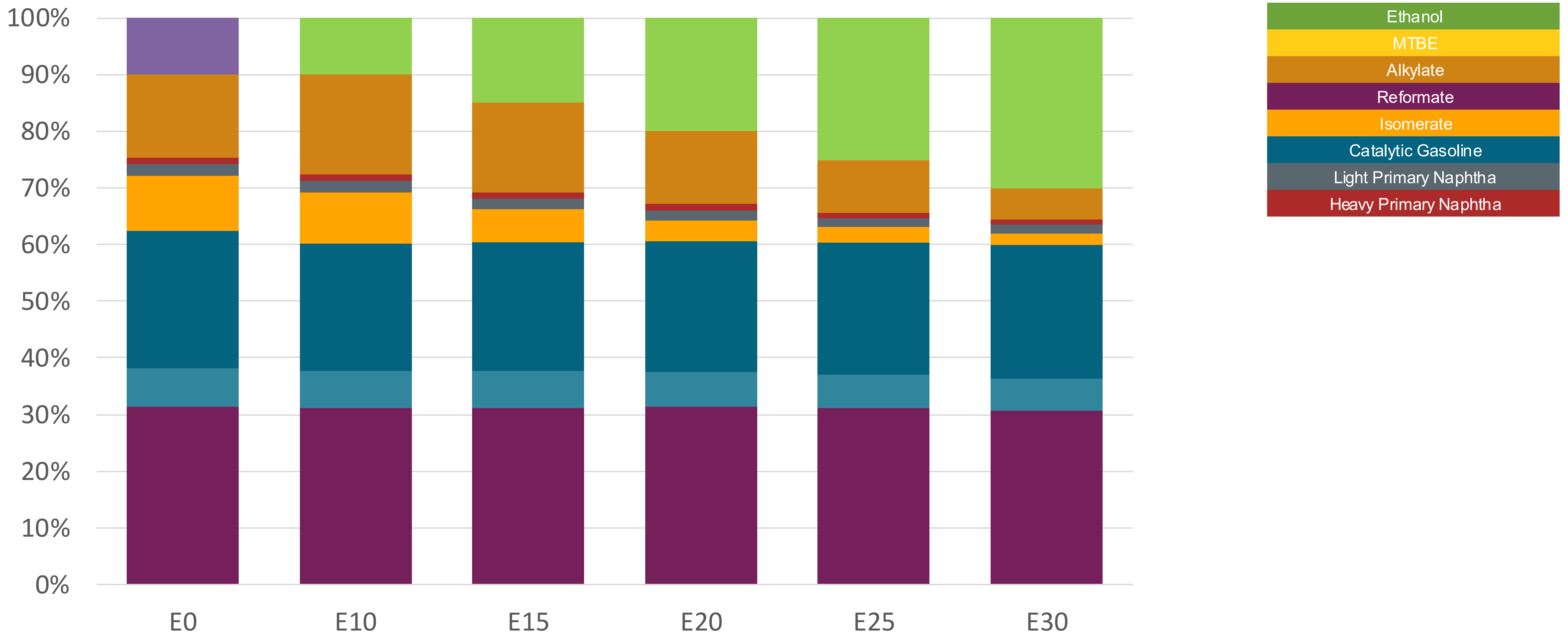
Chile – G93 RM Aconcagua – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	93.0	96.2	98.1	100.0	102.1	104.3
Price (US$/gal)	$ 2.390	$ 2.390	$ 2.390	$ 2.390	$ 2.390	$ 2.390

Source: Faro90, 2025

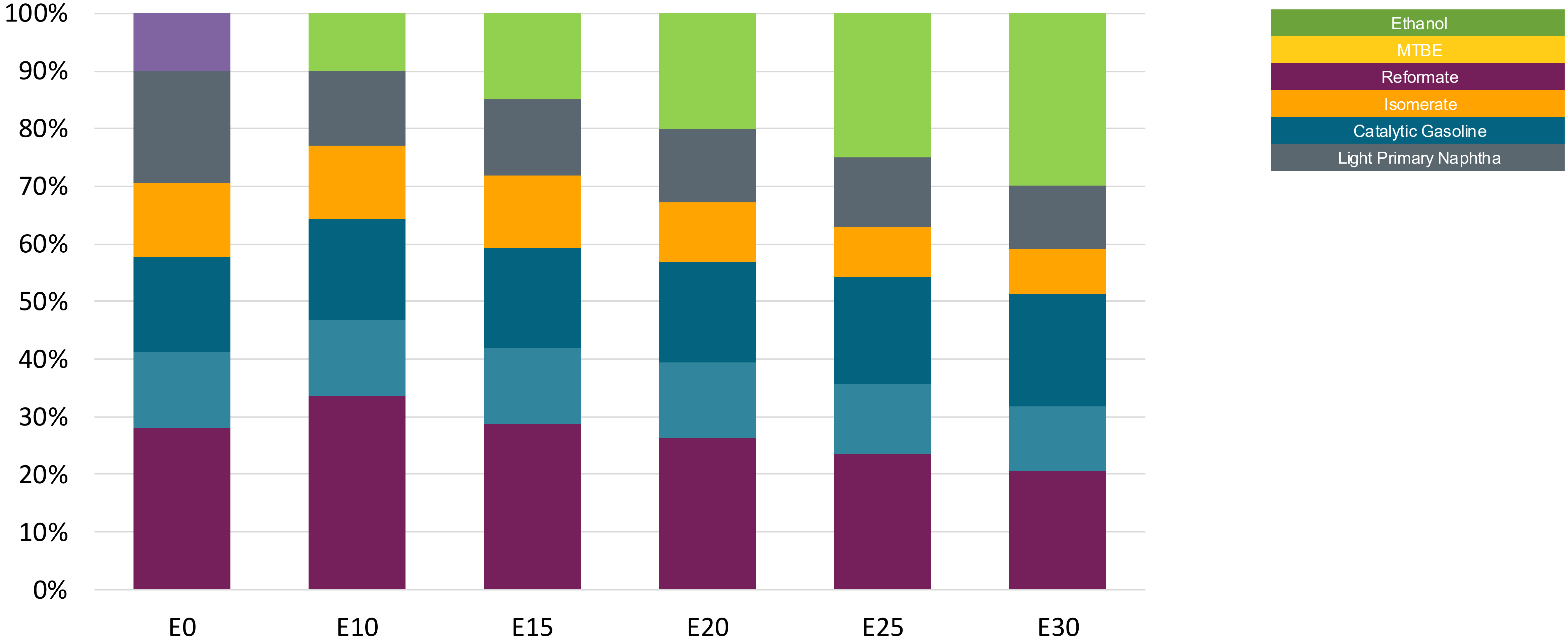
Chile – G97 RM Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.5
Price (US$/gal)	$ 2.52	$ 2.52	$ 2.52	$ 2.52	$ 2.52	$ 2.52

Source: Faro90, 2025

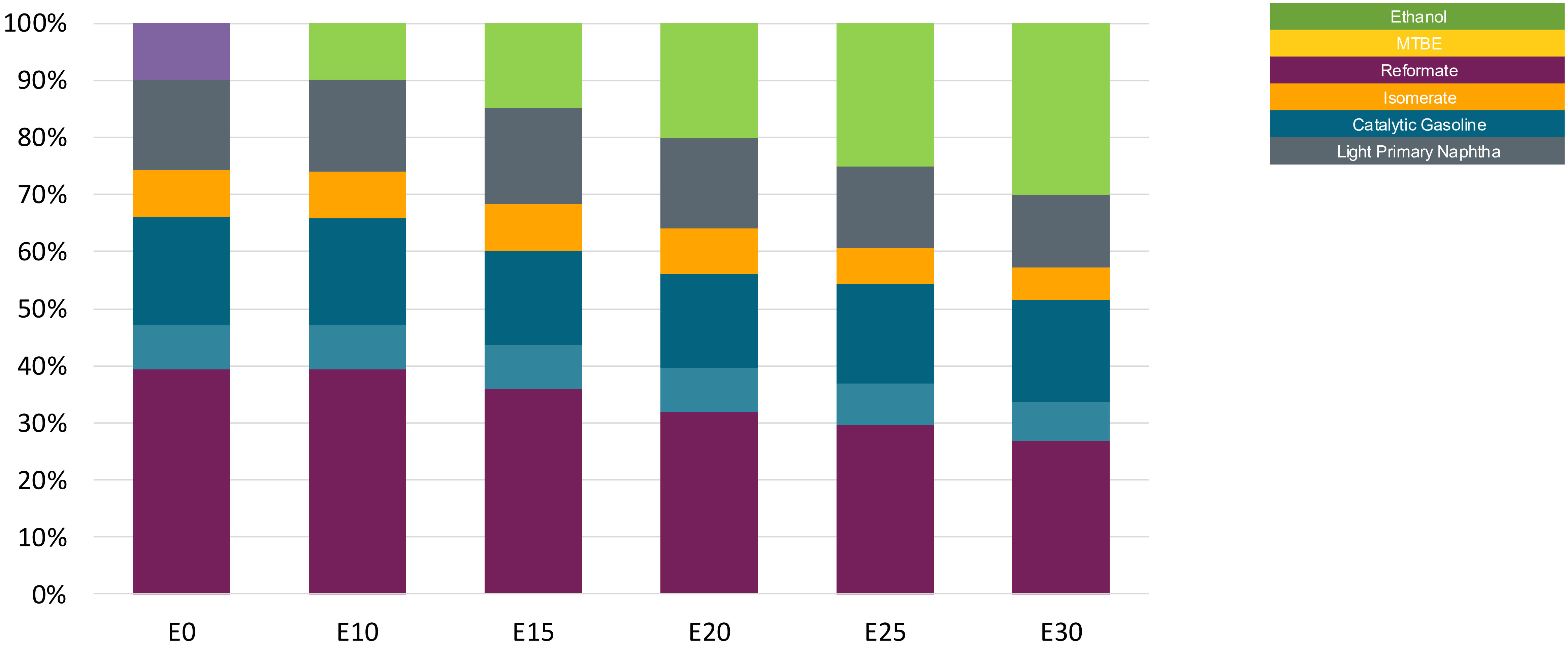
Chile – G93 RP Biobio – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	93.0	93.0	95.9	98.2	100.4	102.7
Price (US$/gal)	$ 2.12	$ 2.12	$ 2.12	$ 2.12	$ 2.12	$ 2.12

Source: Faro90, 2025

Chile – G97 RP Biobío – Octane Increment



Octane (RON)	97.0	101.0	103.7	107.1	108.9	111.1
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Source: Faro90, 2025

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

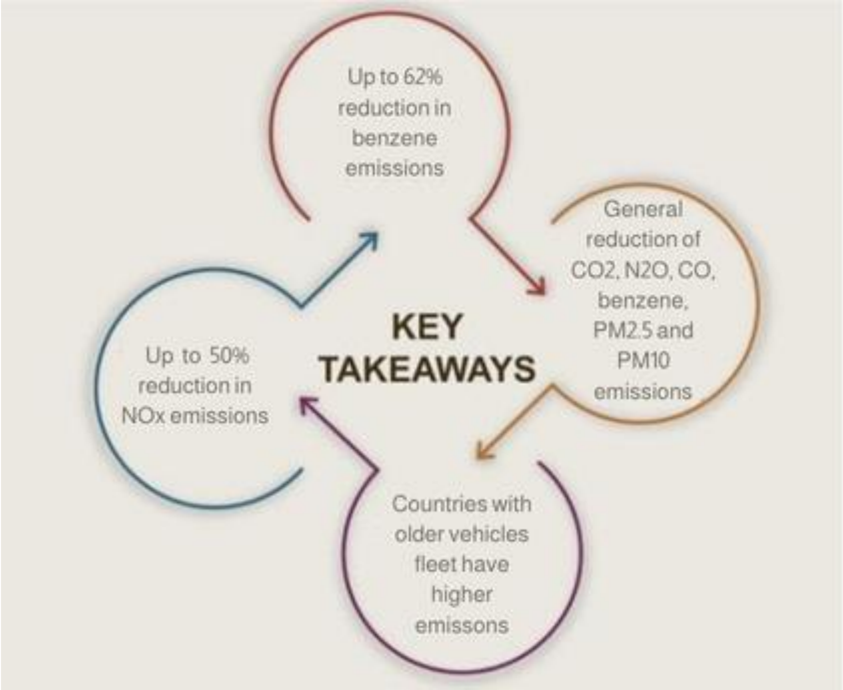
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Chile – Vehicular Fleet

Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	298,823
		Multi FI	3-Ways		>8 years	979,343
			Euro III		>8 years	140,801
			Euro IV		>8 years	772,965
			Euro V		>8 years	413,842
			Euro 6		>8 years	673,001
					4-8 years	803,943
					<4 years	587,674
			Hybrid		All	19,216
		Electric			All	6,303
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	337,941
		Multi FI	3-Ways		>8 years	311,086
			Euro III		>8 years	82,864
			Euro IV		>8 years	228,184
			Euro V		>8 years	157,295
			Euro 6		>8 years	111,611
					4-8 years	315,477
					<4 years	323,360
			Hybrid		All	7,473
		Electric			All	2,451
Sml Engines	Light	4-Cycle	None	PCV	>8 years	31,419
			Euro III		>8 years	2,611
			Euro IV		>8 years	79,444
			Euro V		>8 years	31,326
			Euro 6		>8 years	41,379
					4-8 years	59,001
					<4 years	62,592
			Hybrid		All	0
		Electric			All	0
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	0
		Fuel-Injection	3-Ways		>8 years	165,528
			Euro III		>8 years	12,406
			Euro IV		>8 years	45,582
			Euro V		>8 years	27,253
			Euro 6		>8 years	27,401
					4-8 years	0
					<4 years	50,527
			Hybrid		All	0
		Electric			All	4,107

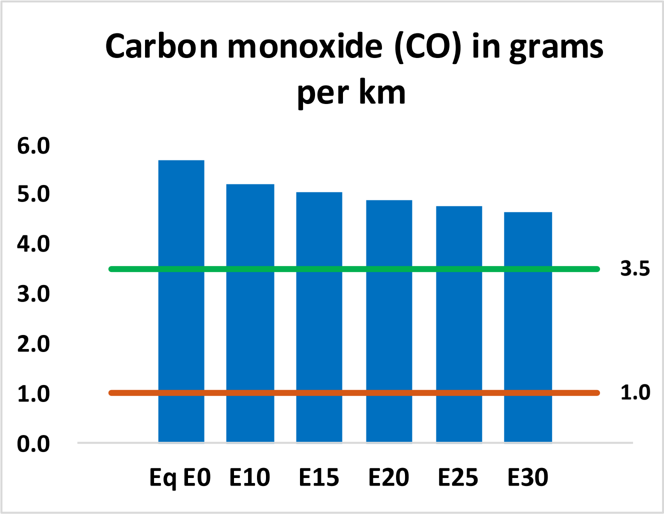
Vehicular Fleet: 6**,818,833 vehicles that use gasoline**
Average Age: **10 year**

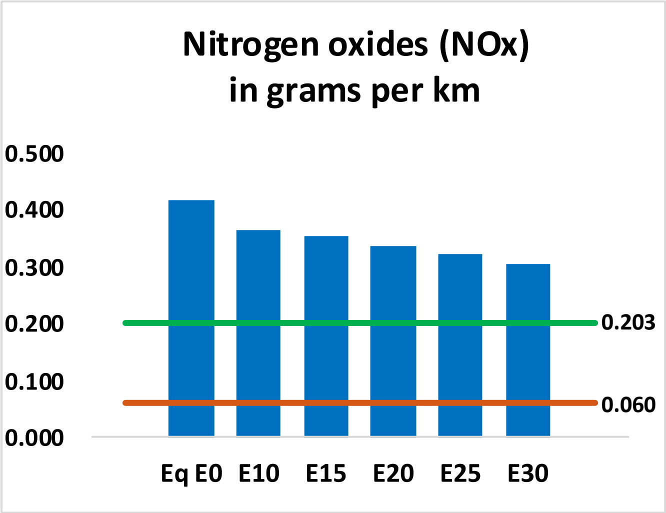
Source: INE, 2024

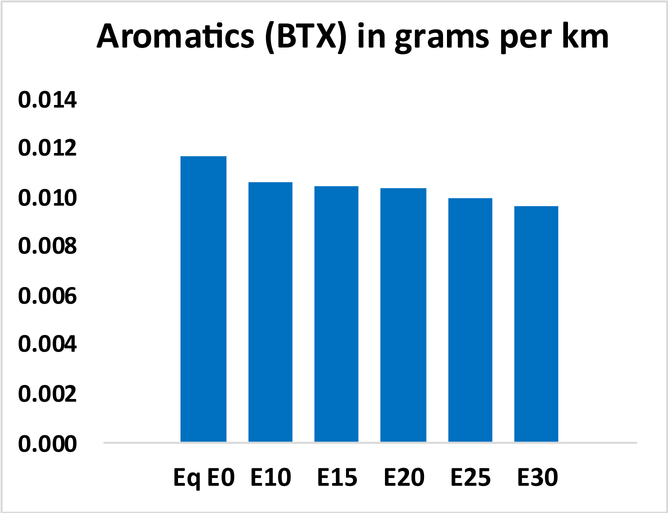
Main Technology: **Motor Multi-Fuel-Injection, Euro V, PCV Tank**

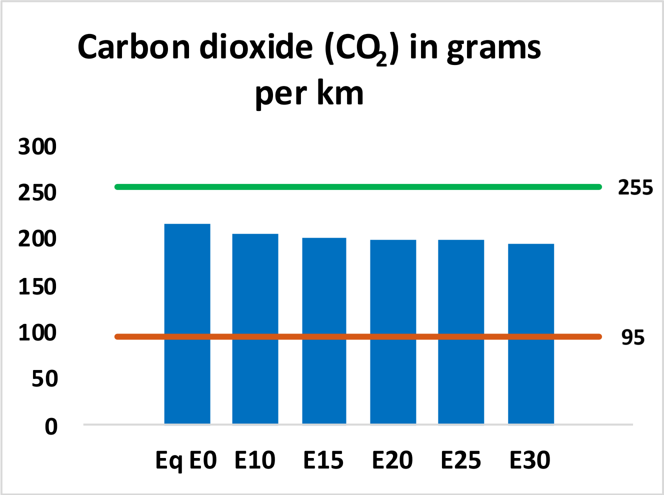
Chile – Gasoline Vehicular Fleet Emissions

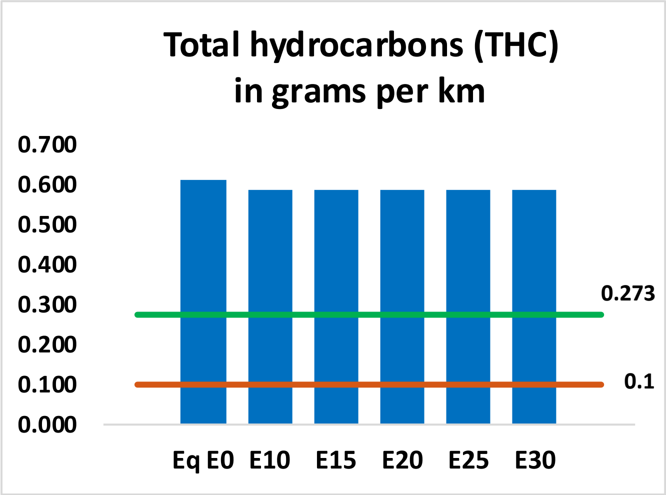
United States
Euro 6

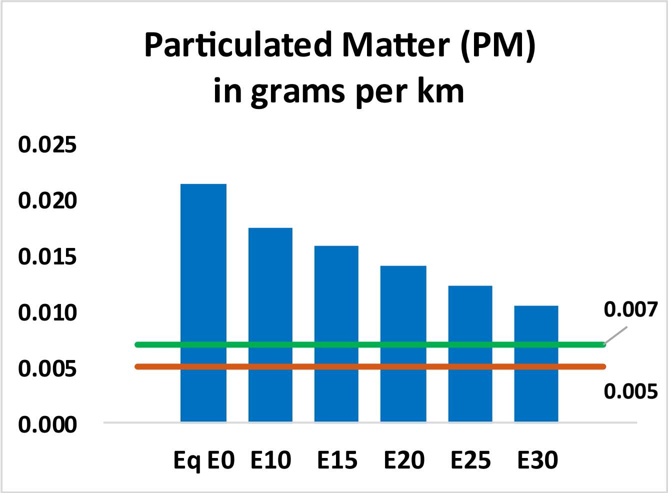












Source: Faro 90

Chile – Gasoline Vehicular Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,022,222	978,171	934,120	904,761	878,317	858,919	831,031	-9%	-14%
CO2	38,858,712	37,859,488	36,915,776	36,173,575	35,806,305	35,604,011	34,947,727	-5%	-8%
VOC	109,513	107,505	105,498	105,105	105,284	105,596	105,399	-4%	-4%
NOx	75,197	69,431	65,538	63,587	60,408	57,933	55,008	-13%	-20%
SOx	446	412	378	350	322	293	262	-15%	-28%
NH3	13,101	12,972	12,842	12,903	13,048	13,150	13,235	-2%	0%
N2O	611	607	605	608	620	627	634	-1%	2%
CH4	24,116	24,116	24,116	24,478	25,009	25,467	25,901	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	687	670	652	645	640	638	631	-5%	-7%
Acetaldehyde	1,363	1,565	2,296	3,497	4,764	5,444	6,432	68%	249%
Formaldehyde	4,080	3,966	4,624	5,429	5,688	6,292	6,850	13%	39%
Benzene	2,100	2,018	1,912	1,882	1,867	1,785	1,727	-9%	-11%
PM2.5	2,240	2,036	1,833	1,654	1,478	1,290	1,094	-18%	-34%
PM10	1,597	1,451	1,306	1,179	1,054	920	779	-18%	-34%

Source: Faro 90

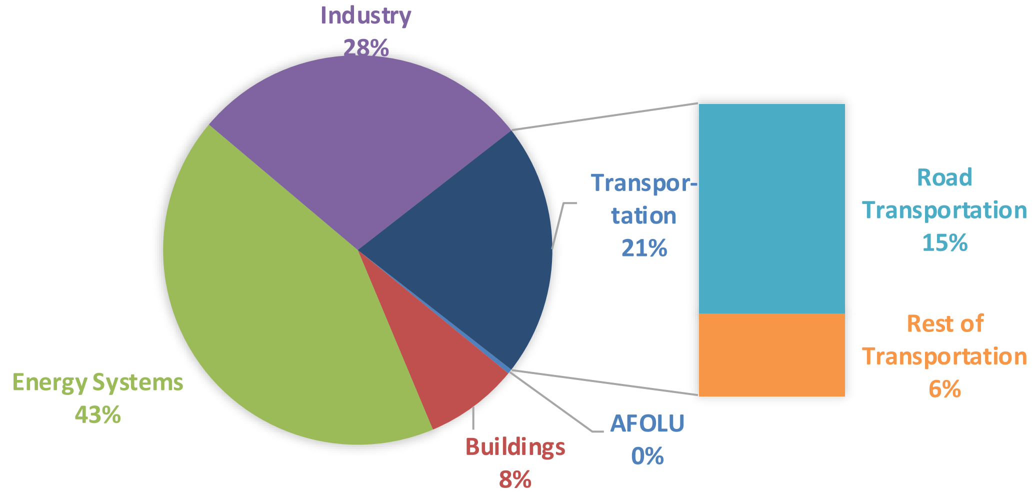
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

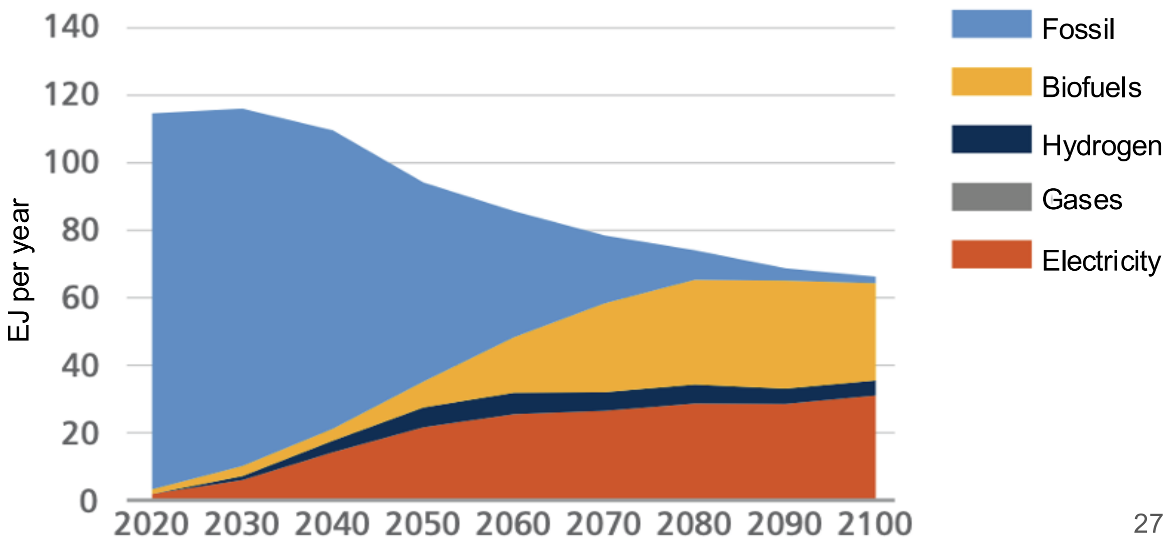
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

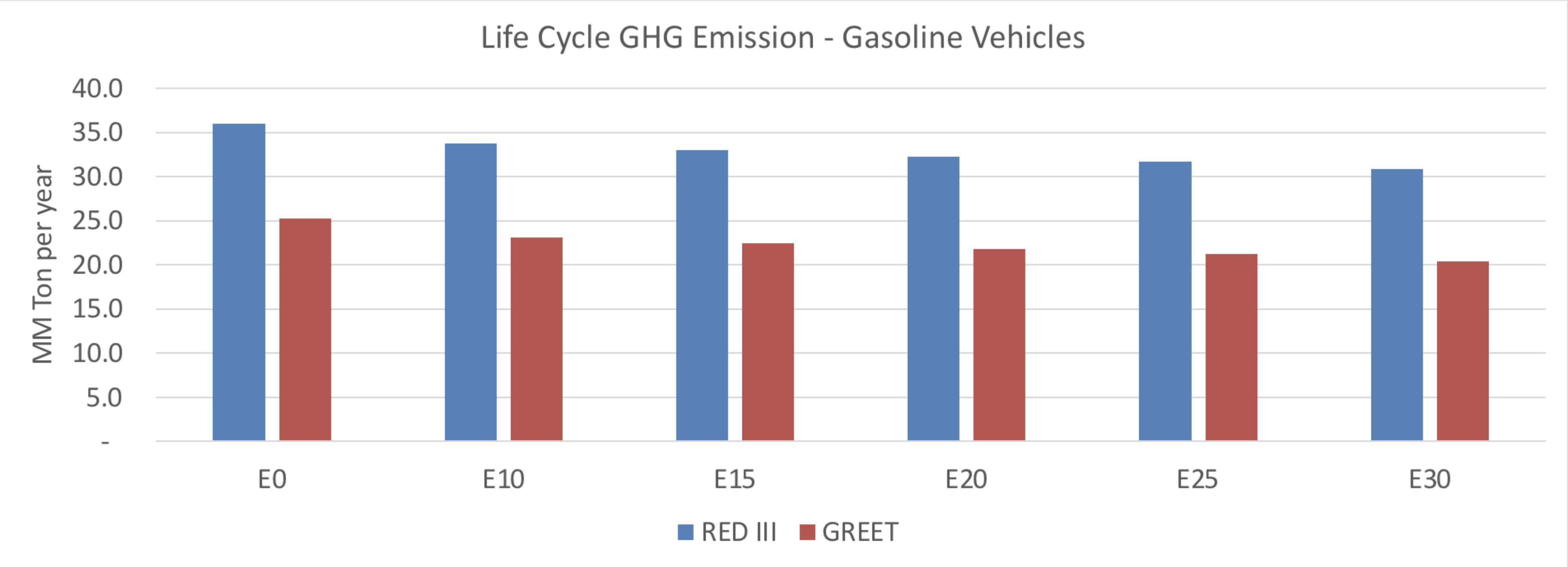
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Chile – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	118.1
2035 Target (MMT/year)	90.1
Delta	28.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.5%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.3%	8.3%	10.3%	12.0%	14.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	7.6%	10.1%	12.5%	14.5%	17.3%
RED II/III	0.0%	8.1%	10.7%	13.3%	15.4%	18.3%

Source: Greet, RED II/III, climateactiontracker.org, F90